

Product Requirements Document

Project: AI Study Assistant Website

1. Product Name

AI Study Assistant

2. Project Purpose

The goal of this project is to build a simple full-stack web application that includes a basic AI agent.

This project will help the team understand how to connect a frontend, backend, database, and AI model in one complete product flow.

The focus is not only on writing code, but also on understanding how AI-powered products are designed, built, connected, and presented.

3. Product Overview

AI Study Assistant is a simple website where users can create learning topics, write notes, and ask an AI assistant questions about those topics.

The AI assistant helps users by:

- Answering questions about a selected topic.
 - Explaining concepts in a simple way.
 - Generating a short quiz.
 - Summarizing user notes.
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4. Target Users

The main users are:

- Students who want help studying.
 - Employees who want to organize learning materials.
 - Users who want to ask questions about technical topics.
 - Users who need simple explanations and summaries.
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5. Project Objectives

By building this project, the team should learn and practice:

- Building a basic frontend.
 - Building backend APIs.
 - Connecting frontend with backend.
 - Storing and retrieving data from a database.
 - Calling an AI model from the backend.
 - Displaying AI responses in the frontend.
 - Structuring a small full-stack AI-powered project.
 - Preparing and presenting a working product demo.
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6. Main Features

Feature 1: User Login

Users should be able to log in using a simple email and password.

For this version, authentication should be simple and does not need advanced security.

Requirements:

- User can register with name, email, and password.
 - User can log in.
 - User can log out.
 - User can only see their own topics and notes.
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Feature 2: Dashboard

After login, the user should see a simple dashboard.

Requirements:

- Show a welcome message.
 - Show a list of created study topics.
 - Add a button to create a new topic.
 - Show the number of notes inside each topic.
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Feature 3: Study Topics

Users should be able to create and manage study topics.

Example topics:

- Introduction to Machine Learning
- Python Basics
- Backend APIs
- Neural Networks

Requirements:

- Create a new topic.
- View all topics.
- Edit topic name.
- Delete topic.

Each topic should include:

- Topic title.
 - Short description.
 - Created date.
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Feature 4: Notes

Inside each topic, users should be able to add notes.

Requirements:

- Add a note to a topic.
- View notes under a topic.
- Edit a note.
- Delete a note.

Each note should include:

- Note title.
 - Note content.
 - Created date.
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Feature 5: AI Assistant

Each topic should include an AI assistant chat.

The user can ask questions related to the topic and their notes.

Example Questions:

- "Explain this topic simply."
- "Summarize my notes."
- "Create 5 quiz questions."
- "What are the most important points?"
- "Give me an example."

Requirements:

- User writes a question.
 - Backend sends the question and related notes to the AI model.
 - AI model returns an answer.
 - Answer is displayed in the chat area.
 - Chat history is shown during the session.
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Feature 6: AI Actions

To keep the project simple, the AI assistant should have 3 clear buttons:

1. **Summarize Notes**
2. **Explain Simply**
3. **Generate Quiz**

Summarize Notes:

The AI should summarize all notes inside the selected topic.

Explain Simply:

The AI should explain the topic in beginner-friendly language.

Generate Quiz:

The AI should generate 5 simple quiz questions with answers.

7. User Flow

Main Flow:

1. User opens the website.
2. User registers or logs in.
3. User sees the dashboard.
4. User creates a study topic.
5. User adds notes inside the topic.

6. User opens the AI assistant.
 7. User asks a question or clicks an AI action button.
 8. Backend sends the request to the AI model.
 9. AI response appears on the screen.
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8. Pages Required

Page 1: Login Page

Contains:

- Email input.
 - Password input.
 - Login button.
 - Link to register page.
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Page 2: Register Page

Contains:

- Name input.
 - Email input.
 - Password input.
 - Register button.
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Page 3: Dashboard Page

Contains:

- Welcome message.
 - List of study topics.
 - Create topic button.
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Page 4: Topic Details Page

Contains:

- Topic title and description.
 - List of notes.
 - Add note button.
 - AI assistant section.
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Page 5: AI Chat Section

This can be part of the Topic Details Page.

Contains:

- Chat input.
 - Send button.
 - AI response area.
 - Buttons for:
 - Summarize Notes
 - Explain Simply
 - Generate Quiz
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9. Suggested Technology Stack

Frontend:

- React.js or Next.js
- Tailwind CSS or simple CSS

Backend:

- FastAPI or Node.js Express

Database:

- SQLite for a simple first version
or
- PostgreSQL if the team is ready

AI Integration:

- OpenAI API, Gemini API, Groq API, or any available company-approved model
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10. Simple Database Design

Users Table

Field	Type	Description
id	Integer	User ID
name	String	User name

Field	Type	Description
email	String	User email
password	String	User password
created_at	Date	Account creation date

Topics Table

Field	Type	Description
id	Integer	Topic ID
user_id	Integer	Owner user ID
title	String	Topic title
description	Text	Topic description
created_at	Date	Topic creation date

Notes Table

Field	Type	Description
id	Integer	Note ID
topic_id	Integer	Related topic ID
title	String	Note title
content	Text	Note content
created_at	Date	Note creation date

Chat Messages Table — Optional

This table is optional. If it feels complicated, chat history can be kept only on the frontend during the session.

Field	Type	Description
id	Integer	Message ID
topic_id	Integer	Related topic ID

Field	Type	Description
role	String	user or assistant
message	Text	Message content
created_at	Date	Message time

11. Backend APIs

Authentication APIs

Method	Endpoint	Description
POST	/auth/register	Register new user
POST	/auth/login	Login user
POST	/auth/logout	Logout user

Topic APIs

Method	Endpoint	Description
GET	/topics	Get all user topics
POST	/topics	Create new topic
GET	/topics/{id}	Get topic details
PUT	/topics/{id}	Update topic
DELETE	/topics/{id}	Delete topic

Note APIs

Method	Endpoint	Description
GET	/topics/{id}/notes	Get notes for topic
POST	/topics/{id}/notes	Add note
PUT	/notes/{id}	Update note
DELETE	/notes/{id}	Delete note

AI APIs

Method	Endpoint	Description
POST	/ai/chat	Ask AI a custom question
POST	/ai/summarize	Summarize topic notes
POST	/ai/explain	Explain topic simply
POST	/ai/quiz	Generate quiz

12. AI Assistant Behavior

The AI assistant should act as a helpful study assistant.

AI Instructions:

The assistant should:

- Use simple language.
- Give clear examples.
- Avoid long complicated answers.
- Focus on the selected topic and notes.
- Help the user understand the topic step by step.

Example Prompt Sent to AI:

"You are a helpful study assistant. The user is learning the topic: [topic title]. These are the user notes: [notes]. Answer the user question in a simple and clear way: [question]."

13. Out of Scope

To keep the project simple, the following features are not required in V1:

- Payment system.
- Admin dashboard.
- File upload.
- Voice assistant.
- Complex permissions.
- Advanced authentication.
- Real-time collaboration.
- Mobile application.
- Advanced DevOps setup.
- Vector database or RAG system.

14. V1 Deliverables

By the end of V1, the team should deliver:

- Working login and register pages.
 - Dashboard page.
 - Topic creation and listing.
 - Notes creation and listing.
 - Simple AI assistant connected to the backend.
 - AI buttons for summarizing, explaining, and quiz generation.
 - Clean demo showing the full user flow.
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15. Suggested Sprint Plan

Sprint 1: Project Setup and Basic Pages

Tasks:

- Create frontend project.
- Create backend project.
- Create database.
- Build login and register pages.
- Build basic dashboard UI.

Deliverable:

- User can open the app and see the basic pages.
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Sprint 2: Topics and Notes

Tasks:

- Create topic APIs.
- Create notes APIs.
- Connect frontend with backend.
- Display topics and notes in the UI.

Deliverable:

- User can create topics and add notes.
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Sprint 3: AI Assistant

Tasks:

- Create AI API endpoints.
- Connect backend to AI model.
- Build AI chat UI.
- Add summarize, explain, and quiz buttons.

Deliverable:

- User can ask the AI assistant questions and receive answers.
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Sprint 4: Testing and Demo Preparation

Tasks:

- Test the full user flow.
- Fix bugs.
- Improve UI.
- Prepare demo script.
- Present final V1 demo.

Deliverable:

- Complete working V1 demo.
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16. Success Criteria

The project is considered successful when:

- A user can register and log in.
 - A user can create a study topic.
 - A user can add notes to the topic.
 - A user can ask the AI assistant a question.
 - The AI assistant gives a useful response.
 - The team can explain the full flow from frontend to backend to database to AI model.
 - The team can present a working demo.
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17. Demo Scenario

The demo should follow this scenario:

1. Register a new user.
 2. Log in.
 3. Create a topic called "Machine Learning Basics."
 4. Add two or three notes about machine learning.
 5. Click "Summarize Notes."
 6. Ask the AI: "Explain supervised learning simply."
 7. Click "Generate Quiz."
 8. Show the generated quiz.
 9. Explain how the frontend, backend, database, and AI model are connected.
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18. Important Notes

This project is not about building a perfect product from the first version.

The main goal is to build a clear working version that shows the complete product flow.

The focus should be on:

- Understanding the product idea.
- Writing clean and simple code.
- Connecting frontend, backend, database, and AI model.
- Explaining the implementation clearly.
- Working together as one team.
- Preparing a clean demo.